SOIL SURVEY
OF THE
PEE DEE RESEARCH AND EDUCATION CENTER
DARLINGTON, SOUTH CAROLINA

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SOIL LEGEND

PEE DEE RESEARCH AND DEVELOPMENT CENTER
DARLINGTON COUNTY, SOUTH CAROLINA

SYMBOL	NAME
BlB	Blanton sand, 0 to 3 percent slopes
BoB	Bonneau sand, 0 to 3 percent slopes
Co	Coxville sandy loam
EnA	Eunola loamy sand, 0 to 2 percent slopes
EnB	Eunola loamy sand, 2 to 6 percent slopes
EuA	Eunola sandy loam, O to 2 percent slopes, depositional
GoA	Goldsboro loamy sand, 0 to 2 percent slopes
Gd.A	Goldsboro loamy sand, 0 to 2 percent slopes, depositional
HnA	Hornsville sandy loam, 0 to 1 percent slopes
HnB	Hornsville sandy loam, 1 to 3 percent slopes
HnB2	Hornsville sandy loam, 1 to 3 percent slopes, eroded
HnC2	Hornsville sandy loam, 3 to 5 percent slopes, eroded
HnD2	Hornsville sandy loam, 5 to 8 percent slopes, eroded
Ly	Lynchburg sandy loam
NbA	Noboco loamy sand, 0 to 1 percent slopes
NbB	Noboco loamy sand, 1 to 3 percent slopes
NcA	Noboco loamy sand, thin surface, 0 to 1 percent slopes
NcB	Noboco loamy sand, thin surface, 1 to 3 percent slopes
NlA	Noboco sandy loam, 0 to 1 percent slopes

NnA	Norfolk loamy sand, thin surface, 1 to 3 percent slopes
NnB	Norfolk loamy sand, thin surface, 1 to 3 percent slopes
NoA	Norfolk loamy sand, 0 to 1 percent slopes
NoB	Norfolk loamy sand, 1 to 3 percent slopes
OcB	Ocilla sand, 0 to 3 percent slopes
Ra	Rains sandy loam
UcB	Uchee sand, 0 to 3 percent slopes
WaB	Wagram sand , 0 to 3 percent slopes

CONVENTIONAL AND SPECIAL SYMBOLS LEGEND

ULTURAL FEATURES

BOUNOARIES

- **National, state or province**
- **County or parish**
- **Minor civil division**
- **Reservation (national forest or park, state forest or park, and large airport)**
- **land grant**
- Limit of soil survey (label)
- **Field sheet matchline and neatline**

AD HOC BOUNDARY (label)

- **Small airport, airfield, park, oilfield, cemetery, or flood pool** — Swift Airport; FLOOD POOL LINE

STATE COORDINATE TICK

LAND DIVISION CORNER (sections and land grants)

ROADS

- Divided (median shown **if scale permits)**
- **·1er roads**
- **.ail**

ROAD EMBLEM & DESIGNATIONS

- **Interstate**
- Federal
- State — ®
- **County, farm or ranch**

RAILROAD

POWER TRANSMISSION LINE **(normally not shown)**

PIPE LINE **(normally not shown)**

FENCE **(normally not shown)**

LEVEES

- **Without road**
- With road
- **With railroad**

DAMS

- Large (to scale)
- **"?dium or Small** — water; w
- **Gravel pit**
- **Mine or quarry**

MISCELLANEOUS CULTURAL FEATURES

- **Farmstead, house (omit in urban areas)**
- Church
- School
- Indian mound (label) — Indian Λ. Mound
- Located object (label) — Tower O
- Tank (label) — Gas
- **Wells, oil or gas**
- Windmill
- **Kitchen midden**

WATER FEATURES

DRAINAGE

- **Perennial, double line**
- **Perennial, single line**
- **Intermittent**
- **Drainage end**
- **Canls or ditches**
  - Double·line (label)
  - **Drainage and/or irrigatiop**

LAKES, PONDS AND RESERVOIRS

- **Perennial** — water; w
- **Intermittent** — int; i

MISCELLANEOUS WATER FEATURES

- **Marsh or swamp**
- Spring
- **Well, artesian**
- **Well, irrigation**
- Wet spot

SPECIAL SYMBOLS FOR SOIL SURVEY

SOIL DELINEATIONS AND SYMBOLS — BoB PaC2

ESCARPMENTS

- Bedrock (points down slope)
- Other than bedrock (points down slope)

SHORT STEEP SLOPE

GULLY

DEPRESSION OR SINK

SOIL SAMPLE (normally not shown)

MISCELLANEOUS

- Blowout
- Clay spot
- **Gravelly spot**
- Gumbo, slick or scabby spot (sodic) — ¢
- **Dumps and other similar non soil areas**
- **Prominent hill or peak**
- **Rock outcrop (includes sandstone and shale)**
- **Saline spot** — +
- Sandy spot
- **Severely eroded spot**
- Slide or slip flips point upslope) —))
- **Stony SPol. ver)· stony spot** — o m
- **Belter drained area ·as much as 2 acres**
- **Water ·as much as 2 acres** — <D
- **Udor1hents and lldipsamments . as much dS 2 acres**

BLANTON SERIES

TAXONOMIC CLASSIFICATION:

Loamy, siliceous, thermic Grossarenic Paleudults .

REPRESENTATIVE PEDON

Ap -- 0 to 7 inches; brown (lOYR 5/3) sand; weak fine granular structure; friable; common fine and medium roots; neutral; abrupt wavy boundary.

El -- 7 to 25 inches; light yellowish brown (lOYR 6/3) sand; weak fine granular structure; very friable; few fine roots; slightly acid; gradual wavy boundary .

E2 -- 25 to 41 inches; pale brown (lOYR 6/3) sand; single grained; loose; few fine roots; slightly acid; gradual wavy boundary .

Btl -- 41 to 50 inches; yellowish brown (lOYR 6/6) sandy clay loam; few distinct clay films on faces of peds; common fine pores; common medium distinct reddish yellow (7.5YR 6/8) masses of iron accumulation; strongly acid; gradual wavy boundary.

Bt2 -- 50 to 61 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; friable; few distinct clay films on faces of peds; common medium prominent reddish yellow (5YR 5/6) and common medium prominent red (2.5YR 4/8) masses of iron accumulation; common medium distinct gray (lOYR 5/1) iron depletions with clear boundaries in the matrix; strongly acid.

Btg -- 61 to 72 inches; gray (lOYR 5/1) sandy clay; weak medium subangular blocky and massive structure; firm; common medium prominent red (2.5YR 4/8) and common medium distinct brownish yellow (lOYR 6/6) masses of iron accumulation; very strongly acid.

RANGE IN CHARACTERISTICS

Solum thickness ranges from 60 to more than 80 inches. Reaction is very strongly acid to slightly acid throughout.

The A or Ap horizon has hue of lOYR, value of 3 to 5, and chroma of 1 to 4. It is sand.

The E horizon has hue of lOYR, value of 4 to 6, and chroma of 2 to 6. It is sand.

The upper part of the Bt horizon has hue of 7.5YR or lOYR, value of 5 to 7, and chroma of 3 to 8. It is sandy loam or sandy clay loam.

The lower part of the Bt horizon has hue of 7.5YR or lOYR, value of 5 to 8 and chroma of 3 to 8 or it is multicolored without a dominant matrix. Iron depletions are commonly within the upper 10 inches of the Bt horizon. It is sandy clay loam.

The Btg horizon has hue of lOYR, value of 6 to 8 and chroma of 1 or 2 or it is dominated by chroma 2 or less with iron accumulations of red, brown and yellow. It is sandy clay loam, or sandy clay.

GEOGRAPHICALLY ASSOCIATED SOILS

These are the Bonneau, Coxville, Noboco, Norfolk, Ocilla, Rains, and Wagram soils. Bonneau, Wagram, Norfolk, Noboco and Ocilla soils have Bt horizons within 40 inches. Coxville and Rains soils have gray horizons immediately below surface layer.

REPRESENTATIVE PEDON LOCATION:

A typical profile of Blanton sand, 0 to 3 percent slopes, is located 0.8 mile northeast from the entrance of the research center. 0.1 mile north on farm road; 0.1 mile west.

MAP UNIT DESCRIPTION

BlB - Blanton sand, 0 to *3 percent slopes*

SETTING

This map unit consists of very deep, well drained soils that formed in loamy marine sediments. These soils are on nearly level to gently sloping low ridges and side slopes on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 7 inches -- brownish loamy sand

SUBSURFACE LAYER:

7 to 25 inches -- brownish loamy sand

25 to 41 inches -- brownish sand

SUBSOIL:

41 to 60 inches -- brownish sandy clay loam with red and gray mottles

60 to 72 inches -- grayish sandy clay with yellow and red mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Bonneau and Wagram soils.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 4.0 to 6.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: moderate

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Draughtiness, low nutrient holding capacity, and soil blowing are the major concerns.

BONNEAU SERIES

TAXONOMIC CLASSIFICATION:

Loamy, siliceous, thermic Arenic Paleudults.

REPRESENTATIVE PEDON

Ap -- 0 to 10 inches; grayish brown (lOYR 5/2) sand; weak fine granular structure; very friable; few very fine roots; few clean sand grains; slightly acid; abrupt broken boundary.

El -- 10 to 19 inches; light yellowish pale brown (lOYR 6/4) sand; weak fine granular structure; very friable; few very fine roots; moderately acid; gradual wavy boundary.

E2 -- 19 to 38 inches; very pale brown (lOYR 7/4) loamy sand; weak fine granular structure; friable; common medium distinct light yellowish brown (lOYR 6/4) masses of iron accumulation; strongly acid; gradual wavy boundary.

Btl -- 38 to 42 inches; yellowish brown (lOYR 5/6) sandy clay loam; weak medium subangular blocky structure; very friable; few very fine roots; common very fine pores; few distinct clay films on faces of peds; faces of some peds coated with stripped sand; moderately acid; gradual wavy boundary.

Bt2 -- 42 to 54 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; friable; few fine pores; many coarse clean sand grains; common medium distinct brown (lOYR 5/3) and common medium prominent red (2.5YR 4/8) masses of iron accumulation; moderately acid; gradual wavy boundary.

Bt3 -- 54 to 60 inches; brownish yellow (lOYR 6/8); sandy clay loam; weak medium subangular blocky structure; friable; few distinct clay films on faces of peds; common clean sand grains; common medium distinct gray (lOYR 5/1) iron depletions with clear boundaries in the matrix; common medium distinct (2.5YR 4/6) masses of iron accumulation; strongly acid; gradual wavy boundary.

RANGE IN CHARACTERISTICS

Solum thickness ranges from 60 to more than 80 inches. Reaction is very strongly acid to moderately acid throughout.

The A or Ap horizon has hue of lOYR, value of 3 to 5, and chroma of 1 to 4. It is sand.

The E horizon has hue of lOYR, value of 4 to 6, and chroma of 2 to 6. It is sand.

The upper part of the Bt horizon has hue of 7.SYR or lOYR, value of 5 to 7, and chroma of 3 to 8. It is sandy loam, or sandy clay loam.

The lower part of the Bt horizon has hue of 7.SYR or lOYR, value of 5 to 8 and chroma of 3 to 8. It commonly has masses of iron accumulation in shades of red or yellow. Iron depletions are at depths between 42 and 60 inches. It is sandy clay loam.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Uchee, Blanton, Coxville, Noboco, Norfolk, Ocilla, Rains, Uchee, and Wagram soils. Blanton soils are grossarenic. Coxville and Rains soils have gray horizons immediately below surface layer. Wagram soils do not have mottles of chroma 2 or less before 60 inches. Norfolk and Noboco soils do not have arenic epipedons. Ocilla soils are somewhat poorly drained.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Bonneau loamy sand, 0 to 3 percent slopes, 0.1 mile northeast from the entrance of the research center; 0.6 mile north on paved road; 0.05 east; 100 feet north.

MAP UNIT DESCRIPTION

Bob -- Bonneau loamy sand, 0 to 3 percent slopes

SETTING

This map unit consists of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level to gently sloping low ridges and side slopes on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER :

0 to 10 inches -- grayish sand

SUBSURFACE LAYERS :

10 to 19 inches brownish sand with brown mottles

19 to 38 inches brownish loamy sand with brown mottles

SUBSOIL :

38 to 42 inches brownish sandy clay loam

42 to 54 inches brownish sandy clay loam with brown and red mottles

54 to 60 inches -- yellowish sandy clay loam with reddish and grayish mottles

60 to 65 inches -- brownish sandy clay loam with reddish and grayish mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Blanton and Wagram soils, and the moderately well drained Ocilla soils. Included similar soils make up less than 15 percent of the map unit.

Dissimilar Soils

These are the well drained Norfolk and Noboco soils. Included dissimilar soils make up O to 10 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 3.5 to 5.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: moderate

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Droughtiness, low nutrient holding capacity, and soil blowing are the major management concerns.

COXVILLE SERIES

TAXONOMIC CLASSIFICATION:

Clayey, kaolinitic, thermic Typic Paleaquults

REPRESENTATIVE PEDON

Ap -- 0 to 10 inches; dark gray (lOYR 4/1) sandy loam; weak fine subangular blocky structure; friable; many fine and very fine roots; few very fine pores; moderately acid; gradual wavy boundary .

Btgl -- 10 to 22 inches; gray (lOYR 5/1) clay; moderate medium subangular blocky structure; friable; few fine and very fine roots; common prominent clay films on faces of peds; few very fine pores; few fine distinct reddish yellow (7.5YR 6/8) masses of iron accumulation; strongly acid; gradual wavy boundary .

Btg2 -- 22 to 30 inches; gray (lOYR 5/1) clay; moderate medium subangular blocky structure; friable; few fine and very fine roots; prominent clay films on faces of peds; few very fine pores; common fine distinct yellowish brown (lOYR 5/8) and few fine prominent red (2.5YR 4/8) masses of iron accumulation; strongly acid; gradual wavy boundary.

Btg3 -- 30 to 50 inches; gray (lOYR 5/1) clay; moderate medium subangular blocky structure; friable; few very fine decayed roots; prominent clay films on faces of peds; few very fine pores; many fine distinct yellowish brown (lOYR 5/8) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btg4 -- 50 to 62 inches; gray (N 6/) clay; moderate medium subangular blocky structure; firm; common prominent clay films on faces of peds; few very fine pores; few clean sand grains; common coarse distinct brownish yellow (lOYR 6/6) and few fine prominent red (2.5YR 4/6) masses of iron accumulation; very strongly acid.

RANGE IN CHARACTERISTICS:

Solum thickness exceeds 60 inches. Reaction ranges from extremely acid to strongly acid throughout.

The Ap horizon has hue of lOYR, value of 2 to 5 and chroma of 1 or 2. It is sandy loam.

The Btg horizon has hue of lOYR, value of 4 to 7, and chroma of 0 to 2. It has few to common masses of iron accumulation in shades of yellow, brown, and red. It is sandy clay, clay loam, or clay.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Goldsboro, Lynchburg, Noboco, Ocilla and Rains soils. Goldsboro, Lynchburg, Noboco and Ocilla soils are better drained. Rains soils are fine-loamy.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Coxville sandy loam is located 0.07 mile north of the entrance of the research center.

MAP UNIT DESCRIPTION

Co -- Coxville sandy loam

SETTING

The Coxville series consists of very deep, poorly drained, slowly permeable soils that formed in clayey marine sediments. They are on nearly level areas and in slightly depressed areas on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 10 inches -- grayish sandy loam

SUBSOIL:

10 to 22 inches	gray clay with yellow mottles
22 to 30 inches	grayish clay with brown and red mottles
30 to 50 inches	grayish clay with brown mottles
50 to 62 inches	grayish clay with yellow and red mottles

MAP UNIT COMPOSITION

Similar Soils

These are the poorly drained Rains soils. Included similar soils make up about 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 0 to 1.0 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: slight

SURFACE RUNOFF: slow to ponded

ORGANIC MATTER CONTENT: moderate

AGRONOMIC INFORMATION:

Wetness is the major management concern.

EUNOLA SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Aquic Hapludults

REPRESENTATIVE PEDON

Ap -- 0 to 7 inches; grayish brown (lOYR 5/3) loamy sand; weak fine subangular blocky structure; very friable; common fine and medium roots; moderately acid; abrupt wavy boundary.

E -- 7 to 17 inches; light yellowish brown (lOYR 6/4) loamy sand; weak fine subangular blocky structure; very friable; moderately acid; clear wavy boundary.

Btl -- 17 to 30 inches; brownish yellow (lOYR 6/6) sandy clay loam; weak medium subangular blocky structure; friable; few fine roots; few fine pores; very strongly acid; gradual wavy boundary.

Bt2 -- 30 to 47 inches; yellowish brown (lOYR 6/6) sandy loam and sandy clay loam; weak medium subangular blocky structure; friable; common medium distinct gray (lOYR 6/1) masses of iron accumulation; few medium distinct brown (7.5YR 5.6) masses of iron accumulation; very strongly acid; gradual wavy boundary .

BCg -- 47 to 60 inchesi gray (lOYR 6/1) sandy loam; massive i friable i common medium faint light gray (lOYR 7/1) iron depletions with clear boundaries in the matrix; very strongly acid.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Eunola loamy sand, 0 to 2 percent slopes, is located 30 feet west from the entrance of the research centeri 0.2 mile north on unpaved roadi 0.3 mile west; 200 feet southi 75 feet west.

The A or Ap horizon has hue of lOYR, value of 3 to 5, and chroma of 1 to 4. It is loamy sand.

The E horizon has hue of lOYR, value of 4 to 7, and chroma of 3 to 4. It is loamy sand.

The upper Bt horizon has hue of lOYR, value of 5 to 7, and chroma of 3 to 8. It is sandy loam or sandy clay loam.

The lower Bt horizon has hue of lOYR, value of 4 to 8, and chrom of 1 to 8, or it is multicolored with no dominant hue. It has few to common masses of iron accumulation in shades of red or yellow. It is sandy clay loam or clay loam.

The BCg horizon has hue of lOYR, value of 4 to 8, and chroma of 1 to 2. Masses of iron accumulation in shades of brown or yellow are present in most pedons. It is sandy loam or sandy clay loam.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Goldsboro, Lynchburg, Noboco, Ocilla and Rains soils. Goldsboro, Lynchburg, Noboco, Ocilla and Rains soils do not have a clay content decrease of greater than 20 percent of the maximum within 60 inches of the soil surface. In addition, Noboco soils are well drained, Lynchburg soils are somewhat poorly drained, and Rains soils are poorly drained.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Eunola loamy sand, 0 to 2 percent slopes, is located 30 feet west from the entrance of the research center; 0.2 mile north on unpaved road; 0.3 mile west; 200 feet south; 75 feet west.

MAP UNIT DESCRIPTION

EnA -- Eunola loamy sand, 0 to *2 percent slopes*

SETTING

The Eunola series consists of very deep, moderately well drained, moderately permeable soils that formed in loamy marine and fluvial sediments on the Coastal Plain. These soils are on nearly level upland areas and stream terraces.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 7 inches -- brownish loamy sand

SUBSURFACE LAYER:

7 to 17 inches brownish loamy sand

SUBSOIL:

17 to 30 inches brownish sandy clay loam

30 to 47 inches brownish sandy clay loam with gray mottles

47 to 60 inches -- grayish sandy loam and loamy sand

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Goldsboro and the somewhat poorly drained Lynchburg soils. Included similar soils make up less than 15 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.0 to 3.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern.

MAP UNIT DESCRIPTION

EnB - Eunola loamy sand, 1 to 3 percent slopes.

SETTING

The Eunola series consists of very deep, moderately well drained, moderately permeable soils that formed in loamy marine and fluvial sediments on the Coastal Plain. These soils are on nearly level upland areas and stream terraces.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 7 inches -- brownish loamy sand

SUBSURFACE LAYER:

7 to 17 inches brownish loamy sand

SUBSOIL:

17 to 30 inches brownish sandy clay loam

30 to 47 inches brownish sandy clay loam with gray mottles

47 to 60 inches -- grayish sandy loam and loamy sand

MAP UNIT COMPOSITION

Included Similar Soils

This is the moderately well drained Hornsville soils. Included soils make up 0 to 10 percent of this unit.

SOIL PROPERTIES:

PERMEABILITY: moderate slow

SEASONAL HIGH WATER TABLE DEPTH: 2.0 to 3.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern.

MAP UNIT DESCRIPTION

EuA -- Eunola sandy loam, 0 to 2 percent slopes, depositional

SETTING

The Eunola series consists of very deep, moderately well drained, moderately permeable soils that formed in loamy marine and fluvial sediments on the Coastal Plain. This phase consists of Eunola soils that have been covered with 8 to 12 inches of loamy spoil material dredged from nearby drainage ditches. These soils are on nearly level upland areas and stream terraces.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 4 inches -- brownish sandy loam

4 to 12 inches -- grayish sandy loam

SUBSOIL:

12 to 28 inches brownish sandy clay loam

28 to 38 inches brownish sandy clay loam with gray mottles

38 to 45 inches -- brownish sandy clay loam with grayish and brownish mottles

45 to 60 inches grayish sandy loam and sandy clay loam with brownish mottles

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Goldsboro and soils and the somewhat poorly drained Lynchburg soils. Included similar soils make up less than 15 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.0 to 3.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern.

GOLDSBORO SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Aquic Paleudults

REPRESENTATIVE PEDON

A -- 0 to 8 inches; brown (lOYR 5/3) loamy sand; weak fine subangular blocky structure; friable; few fine roots; slightly acid; clear wavy boundary .

Btl -- 8 to 28 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; friable; few fine roots; few clean sand grains; common medium faint yellowish brown (lOYR 6/6) masses of iron accumulation; moderately acid; gradual wavy boundary.

Bt2 -- 28 to 38 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; friable; common distinct clay films on faces of peds; few clean sand grains; common medium faint yellowish brown (lOYR 5/8) and common medium distinct (lOYR 6/1) masses of iron accumulation; very strongly acid; gradual wavy boundary .

Btgl -- 38 to 49 inches; gray (lOYR 6/1) sandy clay loam; moderate medium subangular blocky structure; friable;

common distinct clay films on faces of peds; common medium distinct yellowish brown (lOYR 5/4), common medium prominent yellowish red (5YR 5/8), and common medium distinct strong brown (7.5YR 5/6) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btg2 -- 49 to 62 inches; gray (lOYR 5/1) sandy clay loam; moderate medium subangular blocky structure; friable; common distinct clay films on faces of peds; common medium distinct strong brown (7.5YR 5/6) and common medium prominent yellowish red (5YR 5/8) masses of iron accumulation; very strongly acid; gradual wavy boundary.

RANGE IN CHARACTERISTICS:

Solum thickness is more than 60 inches. Reaction is extremely acid to strongly acid throughout.

The Ap horizon has hue of lOYR, value of 4 or 5, and chroma of 2 or 3. It is loamy sand. Some areas that near drainage ditches and roads have a sandy clay loam surface layer.

The upper part of the Bt horizon has hue of lOYR, value of 4 to 6, and chroma of 3 to 8. Masses of iron accumulation in shades of red, yellow, or brown range from none to common. It is sandy clay loam.

The lower part of the Bt horizon has hue of lOYR, value of 4 to 7, and chroma of 3 to 8. Iron depletions range from few to common within a depth of 18 to 30 inches. It is sandy clay loam, or clay loam.

The Btg horizon has hue of lOYR, value of 4 to 7, and chroma of 1 through 8. Masses of iron accumulation in shades of red, yellow, or brown range from few to common. It is sandy clay loam or clay loam

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Bonneau, Eunola, Lynchburg, Noboco, Ocilla, and Rains soils. Eunola soils have a clay content decrease of greater than 20 percent of the maximum within 60 inches of the soil surface. Noboco soils are well drained. Lynchburg soils are somewhat poorly drained. Rains soils are poorly drained. Bonneau and Ocilla soils have A and E horizons more than 20 inches thick.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Goldsboro sandy loam, 0 to 2 percent slopes; 0.5 north from the entrance of the research center; 0.5 mile west on unpaved road; 125 feet south of road.

MAP UNIT DESCRIPTION

GbA -- Goldsboro loamy sand, 0 to *2 percent slopes*

SETTING

These very deep, moderately well drained, moderately permeable soils are on broad level flats on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 8 inches brownish loamy sand

SUBSOIL:

8 to 28 inches -- brownish sandy clay loam with yellow mottles

28 to 38 inches -- brownish sandy clay loam with brown and gray mottles

38 to 62 inches -- grayish sandy clay loam with brown and red mottles

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Eunola and Ocilla soils. The somewhat poorly drained Lynchburg soils and the poorly drained Rains soils. Included similar soils make up less than 15 percent of the map unit .

Dissimilar soils

These are the well drained Noboco and Bonneau soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.0 to 3.0 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern.

MAP UNIT DESCRIPTION

GdA -- Goldsboro sandy clay loam, 0 to *2 percent slopes, depositional*

SETTING

These very deep, moderately well drained, moderately permeable soils on broad flats on the Coastal Plain. This phase consists of Goldsboro soils that have been covered with 8 to 12 inches of loamy spoil material dredged from nearby drainage ditches.

REPRESENTATIVE PEDON

SURFACE LAYERS:

0 to 12 inches - mottled grayish and brownish sandy clay loam

12 to 20 inches, grayish sandy loam

SUBSOIL:

20 to 28 inches - yellowish brown sandy clay loam with brownish yellow mottles

28 to 38 inches - yellowish brown sandy clay loam with yellowish brown and gray mottles

38 to 49 inches - gray sandy clay loam with yellowish brown, strong brown, and yellowish red mottles

49 to 62 inches - gray sandy clay loam with strong brown and yellowish red mottles

62 to 70 inches - gray sandy clay loam with light gray mottles

MAP UNIT COMPOSITION

Similar soils

These are the moderately well drained Eunola and Ocilla soils. The somewhat poorly drained Lynchburg soils and the poorly drained Rains soils. Included similar soils make up less than 15 percent of the map unit.

Dissimilar soils

These are the well drained Noboco and Bonneau soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY : moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.0 to 3.0 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern.

HORNSVILLE SERIES

TAXONOMIC CLASSIFICATION:

Clayey, kaolinitic thermic Aquic Hapludults

REPRESENTATIVE PEDON

Ap -- 0 to 2 inches; very dark gray (lOYR 3/1) sandy loam; weak fine granular structure; friable; many fine and few medium roots; moderately acid; abrupt wavy boundary.

E -- 2 to 4 inches; pale brown (lOYR 6/3) sandy loam; weak fine subangular blocky structure; friable; many fine roots; few fine pores; few quartz gravel; moderately acid; abrupt wavy boundary.

BE -- 4 to 13 inches; yellowish brown (lOYR 5/4) sandy clay loam; common medium prominent yellowish red (5YR 5/8) mottles; weak medium subangular blocky structure; friable; common distinct clay films on faces of peds; few very fine roots; common medium prominent yellowish red (5YR 5/8) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btl -- 13 to 22 inches; yellowish brown (lOYR 5/4) sandy clay; moderate medium subangular blocky structure; friable; few very fine roots; common distinct clay films on faces of peds; common medium distinct yellowish red (5YR 5/8) and common fine distinct gray (lOYR 6/1) iron depletions with clear boundaries in the matrix; very strongly acid; gradual wavy boundary.

Bt2 -- 22 to 27 inches; brownish yellow (lOYR 5/4) clay; strong medium subangular blocky structure; firm; few very fine roots; common distinct clay films on faces of peds; fine distinct yellowish red (5YR 5/8) masses of iron accumulation; common medium distinct gray (lOYR 5/1) iron depletions with clear boundaries in the matrix; very strongly acid; gradual wavy boundary.

Btgl -- 27 to 37 inches; gray (lOYR 5/1) clay, moderate medium subangular blocky structure; firm; common very fine roots; few fine pores; many distinct clay films on faces of peds; common medium distinct yellowish brown (lOYR 5/4) and red (2.5YR 4/8) masses of iron accumulation; very strongly acid; gradual wavy boundary .

Btg2 -- 37 to 46 inches; gray (lOYR 5/1) sandy clay; weak medium subangular blocky structure; firm in place; common medium prominent red (2.5YR 5/8) and common medium distinct yellowish brown (lOYR 5/4) masses of iron accumulation; very strongly acid; gradual wavy boundary.

BCg -- 46 to 60 inches; gray (N 6/) sandy clay and sandy clay loam; massive; friable; pockets of stripped sand; common medium distinct light gray (lOYR 7/2) iron depletions with clear boundaries in the matrix; common medium prominent red (2.SYR 4/8) masses of iron accumulation; ; very strongly acid.

RANGE IN CHARACTERISTICS:

Solum thickness ranges from 40 to more than 60 inches. Reaction is strongly or very strongly acid.

The A or Ap horizon has hue of lOYR, value of 2 to 5, and chroma of 1 to 4. It is sandy loam.

The E horizon has hue of lOYR, value of 4 to 7, and chroma of 1 to 4. It is sandy loam.

The BE horizon has hue of lOYR, value of 4 to 7, and chroma of 1 to 4. It is sandy clay loam

The Bt horizon has hue of SYR to lOYR, value of 4 to 7 and chroma of 3 to 8. Iron depletions occur within 24 inches o the argillic horizon. It is clay loam, sandy clay, or clay.

The lower part of the Bt horizon has hue of 7.5YR or lOYR, value of 4 to 7 and chroma of 1 to 8, or it is multicolored without a dominant hue. It is clay loam, sandy clay, or clay.

The Btg horizon has hue of 7.5YR or lOYR, value of 4 to 7 and chroma of 1 to 3. Few to common masses of iron accumulation in shades of red, and yellow are in most pedons. It is sandy clay, or clay.

The BCg horizon has hue of lOYR, value of 4 to 7 and chroma of 1 to 3. Few to common masses of iron accumulation in shades of red, yellow, and brown are present in some pedons. It is sandy clay, or clay.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the well drained Noboco soils. Noboco soils are fine-loamy.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Hornsville sandy loam, 1 to 3 percent slopes, eroded, is located 0.3 mile north on paved road from the entrance of the research center; 250 feet east of road.

MAP UNIT DESCRIPTION

HnA - Hornsville sandy loam, 0 to *1 percent slopes.*

SETTING

The Hornsville series consists of very deep, moderately well drained, moderately slowly permeable soils that formed in clayey marine sediments. These soils are on nearly level areas on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER :

0 to 10 inches grayish sandy loam

SUBSOIL :

10 to 14 inches - brownish sandy clay loam

14 to 24 inches - brownish sandy clay with grayish mottles

24 to 57 inches - grayish clay with yellowish and reddish mottles

57 to 65 inches - grayish and reddish sandy clay loam, sandy clay, and clay

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Eunola soils. Also included are small areas of Hornsville soils that have been covered by more than 10 inches of spoil material dredged from drainage ditches. Included similar soils make up less than 10 percent of the unit.

Dissimilar Soils

These are the poorly drained Coxville and Rains soils. These included soils make up 0 to 10 percent of the unit.

SOIL PROPERTIES

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 3.5 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern for this soil.

<u>MAP UNIT DESCRIPTION</u>

HnB - Hornsville sandy loam, 1 to 3 percent slopes.

SETTING

The Hornsville series consists of very deep, moderately well drained, moderately slowly permeable soils that formed in clayey marine sediments. These soils are on gently sloping areas on the Coastal Plain.

<u>REPRESENTATIVE PEDON</u>

SURFACE LAYER:

0 to 10 inches - grayish sandy loam

SUBSOIL:

10 to 14 inches - brownish sandy clay loam

14 to 24 inches - brownish sandy clay with grayish mottles

24 to 57 inches - grayish clay with yellowish and reddish mottles

57 to 65 inches - grayish and reddish sandy clay loam, sandy clay, and clay

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Eunola. Included similar soils make up less than 15 percent of the unit.

<u>SOIL PROPERTIES</u>

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 3.5 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: moderate

SURFACE RUNOFF: moderate

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

MAP UNIT DESCRIPTION

HnB2 - Hornsville sandy loam, 1 to 3 percent slopes, eroded.

SETTING

The Hornsville series consists of very deep, moderately well drained, moderately slowly permeable soils that formed in clayey marine sediments. These soils are on gently sloping areas on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER :

0 to 2 inches - grayish sandy loam

SUBSURFACE LAY ER :

2 to 4 inches - brownish sandy loam

SUBSOI L :

4 to 13 inches - brownish sandy clay with reddish mottles

13 to 27 inches - brownish sandy clay and clay with reddish and grayish mottles

27 to 46 inches - grayish clay and sandy clay with reddish and brownish mottles

SUBST RATUM :

46 to 60 inches - grayish sandy clay loam and sandy clay

These are the moderately well Eunola soils on similar landscapes. Included similar soils make up less than 15 percent of the unit.

Dissimilar Soils

These are the poorly drained Coxville and Rains soils are on lower landscapes. These included soils make up 0 to 10 percent of the unit.

SOIL PROPERTIES

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 3.5 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: moderate

SURFACE RUNOFF: moderate

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

<u>MAP UNIT DESCRIPTION</u>

HnC2 - Honsville sandy loam, 3 to 5 *percent slopes, eroded.*

SETTING

The Hornsville series consists of very deep, moderately well drained, moderately slowly permeable soils that formed in clayey marine sediments. These soils are on gently sloping areas on the Coastal Plain.

<u>REPRESENTATIVE PEDON</u>

SURFACE LAYER :

0 to 2 inches - grayish sandy loam

SUBSURFACE LAYER :

2 to 4 inches - brownish sandy loam

SUBSOIL :

4 to 13 inches - brownish sandy clay with reddish mottles

13 to 27 inches - brownish sandy clay and clay with reddish and grayish mottles

27 to 46 inches - grayish clay and sandy clay with reddish and brownish mottles

SUBSTRATUM :

46 to 60 inches - grayish sandy clay loam and sandy clay

These are the well drained Eunola soils on similar landscapes. Included similar soils make up less than 15 percent of the unit.

Dissimilar Soils

These are the poorly drained Coxville and Rains soils are on lower landscapes. These included soils make up 0 to 10 percent of the unit.

SOIL PROPERTIES

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 3.5 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: moderate

SURFACE RUNOFF: moderate

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

MAP UNIT DESCRIPTION

HnD2 - Hornsville sandy loam, 1 to 3 percent slopes, eroded.

SETTING

The Hornsville series consists of very deep, moderately well drained, moderately slowly permeable soils that formed in clayey marine sediments. These soils are on gently sloping areas on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 2 inches - grayish sandy loam

SUBSURFACE LAYER:

2 to 4 inches - brownish sandy loam

SUBSOIL:

4 to 13 inches - brownish sandy clay with reddish mottles

13 to 27 inches - brownish sandy clay and 6lay with reddish and grayish mottles

27 to 46 inches - grayish clay and sandy clay with reddish and brownish mottles

SUBSTRATUM:

46 to 60 inches - grayish sandy clay loam and sandy clay

These are the well drained Eunola soils on similar landscapes. Included similar soils make up less than 15 percent of the unit.

Dissimilar Soils

These are the poorly drained Coxville and Rains soils are on lower landscapes. These included soils make up 0 to 10 percent of the unit.

SOIL PROPERTIES

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 3.5 feet

AVAILABLE WATER CAPACITY: high

EROSION HAZARD: moderate

SURFACE RUNOFF: moderate

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

LYNCHBURG SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Aerie Paleaquults

REPRESENTATIVE PEDON

Ap -- O to 11 inches; grayish brown (lOYR 5/2) sandy loam; weak fine subangular blocky structure; very friable; common medium and fine roots; few small pockets of loamy B material; few clean sand grains; strongly acid; clear wavy boundary.

BE -- 11 to 15 inches; pale brown (lOYR 6/3) sandy loam; weak medium subangular blocky structure; friable; few fine roots; common medium distinct light gray (lOYR 7/2) iron depletions with clear boundaries in the matrix; and few medium distinct brownish yellow (lOYR 6/8) masses of iron accumulation; moderately acid; clear wavy boundary.

Btgl -- 15 to 30 inches; light brownish gray (lOYR 6/2) sandy clay loam; weak medium subangular blocky structure; friable; few medium distinct clay films on faces of peds; common fine pores; common medium distinct brownish yellow (lOYR 6/8) masses of iron accumulation and common medium prominent yellowish red (5YR 5/8) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btg2 -- 30 to 49 inches; light brownish gray (lOYR 6/1) sandy clay loam; weak medium subangular blocky structure; friable; few medium distinct clay films on faces of peds; common fine pores; common medium distinct yellowish brown (lOYR 5/6) masses of iron accumulation with clear in the matrix and common medium prominent yellowish red (5YR 5/8) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btg2 -- 49 to 75 inches; gray (lOYR 6/1) sandy clay loam; weak medium subangular blocky structure; friable; common distinct clay films on faces of peds; common fine pores; common medium distinct brownish yellow (lOYR 5/6) masses of iron accumulation; very strongly acid; gradual wavy boundary.

<u>RANGE IN CHARACTERISTICS:</u>

Solum thickness exceeds 60 inches. Reaction. is extremely acid to strongly acid throughout .

The Ap horizon has hue of lOYR, value of 2 to 4, and chroma of 0 to 2 . It is sandy loam.

The BE horizon has hue of lOYR or 2.5Y, value of 5 to 6 and chroma of 3 to 6. It is sandy loam.

The Bt horizon has hue of lOYR or 2.5Y, value of 5 to 6, and chroma of 3 to 8. It is sandy loam, sandy clay loam, loam, or clay loam.

The Btg horizon has hue of lOYR to 2.5Y, value of 4 to 7, and chroma of 0 to 2. It is loam, sandy clay loam, or clay loam.

<u>GEOGRAPHICALLY ASSOCIATED SOILS:</u>
These are the Coxville, Eunola, Goldsboro, Ocilla and Rains soils. Ocilla soils have A and E horizons more than 20 inches thick. Coxville and Rains soils have dominant chroma of 2 or less throughout the horizon. Goldsboro soils do not have gray ottles of chroma 2 or less within 18 inches of the surface. Eunola soils decrease in clay content by more than 20 percent of the maximum within 60 inches of the surface.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Lynchburg sandy loam, 0 to 2 percent slopes, is located 0.5 mile north on paved road from the entrance of the research center; 0.3 mile west on unpaved road; 200 feet south west of road.

MAP UNIT DESCRIPTION

Ly -- Lynchburg sandy loam, 0 to *2 percent slopes.*

SETTING

These very deep, somewhat poorly drained, moderately permeable soils formed in loamy marine sediments. They are on broad flats on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

10 to 11 inches - grayish loamy sand

SUBSOIL:

11 to 15 inches - brownish sandy clay loam with grayish mottles

15 to 75 inches - grayish sandy clay loam with yellowish and reddish mottles

MAP UNIT COMPOSITION

Similar soils

These are the moderately well drained Goldsboro and Ocilla soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

The poorly drained Rains and Coxville soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE: 0.5 to 1.5 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF : slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness is the major management concern .

NOBOCO SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Typic Paleudults

REPRESENTATIVE PEDON

Ap -- 0 to 7 inches; dark grayish brown (lOYR 4/2) loamy sand; weak fine subangular blocky structure; very friable; few very fine and fine roots; few clean sand grains; few pockets of E horizon material; neutral; abrupt wavy boundary.

E -- 7 to 13 inches; very pale brown (lOYR 7/4) loamy sand; weak fine subangular blocky structure; very friable; few very fine roots; neutral; gradual wavy boundary .

Btl -- 13 to 31 inches; yellowish brown (lOYR 5/8) sandy clay loam; moderate medium subangular blocky structure; friable; few very fine roots; common fine pores; distinct clay films on faces of peds and along old root channels; few clean sand grains; strongly acid; gradual wavy boundary.

Bt2 -- 31 to 45 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium faint yellowish brown (lOYR 5/4) and moderate medium distinct yellowish red (5YR 5/8) mottles; moderate medium subangular blocky structure; friable; few very fine pores; few distinct clay films on faces of peds; 2 percent nodules of plinthite; few clean sand grains; common medium faint yellowish brown (lOYR 5/4) and common medium prominent yellowish red (5YR 5/8) masses of iron accumulation;av strongly acid; gradual wavy boundary .

Bt3 -- 45 to 60 inches; strong brown (7.5YR 5/8) sandy clay loam; weak medium subangular blocky structure; friable; common clean sand grains; common medium prominent red (2.5YR 4/6) masses of iron accumulation; common medium distinct gray (lOYR 6/1) iron depletions with clear boundaries in the matrix; very strongly acid .

RANGE IN CHARACTERISTICS:

Solum thickness is more than 60 inches. Reaction is very strongly acid or strongly acid throughout .

The Ap horizon has hue of lOYR, value of 4 to 6, and chroma of 1 to 4 . It is loamy sand .

The E horizon has hue of lOYR, value of 6 or 7, and chroma of 2 to 4 . It is loamy sand.

The upper part of the Bt horizon has hue of 7.5YR or lOYR, value of 3 or 6, and chroma of 3 or 8 . Masses of iron accumulation in shades of red, yellow, or brown range from none to common. It is sandy loam or sandy clay loam.

The lower part of the Bt horizon has hue of 7.SYR or lOYR, value of 5 or 6, and chroma of 3 to 8. Masses of iron accumulation in shades of red, yellow, or brown range from few to common. Iron depletions are within 30 to 48 inches of the surface. It is sandy clay loam or clay loam.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Blanton, Bonneau, Coxville, Goldsboro, Lynchburg, Norfolk, Ocilla and Rains soils. Blanton, Bonneau, and Ocilla soils have A and E horizons more than 20 inches thick. Coxville and Rains have dominant chroma of 2 or less throughout the horizon. Goldsboro and Lynchburg soils have gray mottles of chroma 2 or less within 30 inches of the surface. Norfolk soils do not have gray mottles of chroma 2 or less within 48 inches of the surface.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Noboco loamy sand, 0 to 2 percent slopes, is located 0.5 mile north on paved road from the entrance of the research center; 0.3 mile west on unpaved road; 0.3 mile south on unpaved road; 200 feet west.

<u>MAP UNIT DESCRIPTION:</u>

NbA -- Noboco loamy sand, 0 to *1 percent slopes.*

SETTING

The Noboco series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level upland areas on the Coastal Plain.

<u>REPRESENTATIVE PEDON</u>

SURFACE LAYER:

0 to 7 inches -- brownish loamy sand

SUBSURFACE LAYER:

7 to 13 inches brownish loamy sand

SUBSOIL:

13 to 31 inches brownish sandy. clay loam

31 to 45 inches brownish sandy clay loam with yellow and red mottles

45 to 60 inches -- brown sandy clay loam with gray and red mottles

<u>MAP UNIT COMPOSITION</u>

Similar Soils

The well drained Norfolk soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau soils and the moderately well drained Goldsboro soils. Included dissimilar soils make up 0 to 10 percent of the map unit .

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 4.0 ft

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

There are no major management concerns.

MAP UNIT DESCRIPTION

NbB -- Noboco loamy sand, 1 to *3 percent slopes.*

SETTING

The Noboco series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on gently sloping upland areas on the Coastal Plain.

REPRESENTATIVE PROFILE

SURFACE LAYER:

0 to 7 inches -- brownish loamy sand

SUBSURFACE LAYER:

7 to 13 inches brownish loamy sand

SUBSOIL:

13 to 31 inches brownish sandy clay loam

31 to 45 inches brownish sandy clay loam with yellow and red mottles

45 to 60 inches -- brownish sandy clay loam with gray and red mottles

MAP UNIT COMPOSITION

Similar Soils

The well drained Norfolk soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau soils and the moderately well drained Goldsboro soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 4.0 ft

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: moderate

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

MAP UNIT DESCRIPTION

SETTING

NcA -- Noboco loamy sand, thin surface, 0 to *1 percent slopes.*

SETTING

The Noboco series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level upland areas on the Coastal Plain. These soils have surface layers less than 10 inches thick.

REPRESENTATIVE

PEDON SURFACE LAYER:

0 to 6 inches -- brownish loamy sand

SUBSURFACE LAYER

6 to 8 inches brownish loamy sand

SUBSOIL:

8 to 31 inches -- brownish sandy clay loam

31 to 48 inches -- brownish sandy clay loam with red and gray mottles

48 to 60 inches -- brownish sandy clay loam with gray and red mottles

Similar Soils

The well drained Norfolk soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau soils and the moderately well. drained Goldsboro soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 4.0 ft

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

There are no major management concerns.

MAP UNIT DESCRIPTION

NcB -- Noboco loamy sand, thin surface, 1 to *3 percent slopes.*

SETTING

The Noboco series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on gently sloping upland areas on the Coastal Plain.

REPRESENTATIVE

PROFILE SURFACE LAYER:

0 to 6 inches -- brownish loamy sand

SUBSURFACE LAYER

6 to 8 inches brownish loamy sand

SUBSOIL:

8 to 31 inches -- brownish sandy clay loam

31 to 48 inches -- brownish sandy clay loam with red and gray mottles

48 to 60 inches -- brownish sandy clay loam with gray and red mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Norfolk and moderately well drained Goldsboro soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau soils and the moderately well drained Goldsboro soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 4.0 ft

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: moderate

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern for this soil.

MAP UNIT DESCRIPTION

NlA - Noboco sandy loam, 0 to *1 percent slopes.*

SETTING

This map unit consists of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level upland areas on the Coastal Plain.

<u>REPRESENTATIVE PEDON</u>

SURFACE LAYER:

0 to 10 inches brownish sandy loam

SUBSURFACE LAYER:

10 to 14 inches -- brownish sandy loam

SUBSOIL:

14 to 32 inches -- yellowish sandy clay loam with brown mottles

32 to 42 inches -- yellowish sandy lay loam with red mottles and gray mottles

42 to 57 inches -- mottled grayish and brownish sandy clay loam

57 to 65 inches -- grayish sandy clay with brown mottles

<u>MAP UNIT COMPOSITION</u>

Similar Soils

The well drained Norfolk soils. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau soils and the moderately well drained Goldsboro soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 2.5 to 4.0 ft

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

There are no major management concerns.

NORFOLK SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Typic Kandiudults

REPRESENTATIVE PEDON

Ap -- 0 to 5 inches; grayish brown (lOYR 5/2) loamy sand; weak fine subangular blocky structure; friable; common very fine, fine and medium roots; few fine pores; common clean sand grains; moderately acid; abrupt wavy boundary.

E -- 5 to 17 inches; light yellowish brown (lOYR 6/4) loamy sand; weak fine subangular blocky structure; friable; few very fine roots; few very fine pores; moderately acid; gradual wavy boundary.

Btl -- 17 to 23 inches; yellowish brown (lOYR 5/8) sandy clay loam; weak medium subangular blocky structure; friable; few very fine roots; common distinct clay films on faces of peds; strongly acid; gradual wavy boundary.

Bt2 -- 23 to 34 inches; yellowish brown (lOYR 5/6) sandy clay loam; weak medium subangular blocky structure;

friable; few fine roots; few fine pores; common distinct clay films on faces of peds; few clean sand grains; very strongly acid; gradual wavy boundary.

Bt3 -- 34 to 54 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; friable; few fine roots; few fine pores; common distinct clay films on faces of peds; few clean sand grains; common medium prominent yellowish red (5YR 5/8) and few medium faint yellowish brown (lOYR 5/4) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Bt4 -- 54 to 63 inches; yellowish brown (lOYR 5/6) sandy clay loam; moderate medium subangular blocky structure; gray part is firm; red and brown parts are friable; common distinct clay films on faces of peds; common fine prominent red (2.5YR 4/8) masses of iron accumulation; common fine distinct gray (lOYR 6/1) iron depletions with clear boundaries in the matrix; very strongly acid.

RANGE IN CHARACTERISTICS

Solum thickness exceeds 60 inches. Reaction is extremely acid to strongly acid throughout. Some pedons have up to 4 percent plinthite .

The Ap horizon has hue of lOYR, value of 5 or 6, and chroma of 2 to 4. It is loamy sand.

The E horizon has hue of lOYR, value of 6 or 7, and chroma of 2 or 6. It is loamy sand.

The upper Bt horizon has hue of 7.5YR or lOYR, value of 5 or 6, and chroma of 6 or 8. Masses of iron accumulation in shades of red, yellow or brown range from none to common. It is sandy loam, or sandy clay loam.

The lower Bt horizon has hue of 7.5YR or lOYR, value of 5 or 6, and chroma of 3 to 8. Masses of iron accumulation in shades of red, yellow or brown range from none to common. Iron depletions are within 48 to 72 inches of the surface. It is sandy clay loam, or clay loam.

GEOGRAPHICALLY ASSOCIATED SOILS

These are the Blanton, Bonneau, Coxville, Goldsboro, Lynchburg, Noboco, Rains and Wagram soils. Blanton, Bonneau, and Wagram soils have A and E horizons more than 20 inches thick. Coxville and Rains have dominant chroma of 2 or less throughout the horizon. Goldsboro and Rains soils have gray mottles of chroma 2 or less within 30 inches of the surface. Lynchburg soils have gray mottles of chroma 2 or less in the uppermost layer of the argillic horizon.

Noboco soils have gray mottles of chroma 2 or less within 48 inches of the surface.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Norfolk loamy sand, 0 to 3 percent slopes, is located 0.7 mile north on paved road from the entrance of the research center 0.1 mile east on farm road; 0.2 mile north on farm road; 0.1 mile east on farm road; 600 feet south.

MAP UNIT DESCRIPTION

NnA -- Norfolk loamy sand, thin surface, 0 to *1 percent slopes.*

SETTING

The Norfolk series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridge tops on the Coastal Plain. These soils have surface layers less than 10 inches thick .

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 8 inches   brownish loamy sand

SUBSOIL:

8 to 35 inches -- brownish sandy clay loam

35 to 56 inches -- brownish sandy clay loam with red and brown mottles

56 to 60 inches -- brownish sandy clay with red and gray mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Noboco. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Wagram and Bonneau soils

Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 4.0 to 6 feet

AVAILABLE' WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

There are no major management concerns.

NnB -- Norfolk loamy sand, thin surface, 1 to 3 percent slopes.

SETTING

The Norfolk series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridge tops and side slopes on the Coastal Plain. These soils have surface layers less than 10 inches thick.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 8 inches brownish loamy sand

SUBSOIL:

8 to 35 inches -- brownish sandy clay loam

35 to 56 inches -- brownish sandy clay loam with red and brown mottles

56 to 60 inches -- brownish sandy clay with red and gray mottles

MAP UNIT COMPOSITION

Similar Soils

The well drained Noboco. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Wagram and Bonneau soils

Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 4.0 to 6 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern.

MAP UNIT DESCRIPTION

NoA -- Norfolk loamy sand, 0 to *1 percent slopes.*

SETTING

The Norfolk series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridge tops on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER :

0 to 5 inches -- brownish loamy sand

SUBSURFACE LAYER :

5 to 17 inches brownish loamy sand

SUBSOIL :

17 to 34 inches brownish sandy clay loam

34 to 54 inches brownish sandy clay loam with red and brown mottles

54 to 63 inches -- brownish sandy clay loam with red and gray mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Noboco. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Wagram and Bonneau soils Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASOIJAL HIGH WATER TABLE DEPTH: 4.0 to 6 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

There are no major management concerns .

NoB -- Norfolk loamy sand 1 to *3 percent slopes.*

SETTING

The Norfolk series consist of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridge tops and side slopes on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER :

0 to 5 inches -- brownish loamy sand

SUBSURFACE LAYER :

5 to 17 inches brownish loamy sand

SUBSOIL :

17 to 23 inches brownish sandy clay loam

23 to 34 inches brownish sandy clay loam

34 to 54 inches brownish sandy clay loam with red and brown mottles

54 to 63 inches -- brownish sandy clay loam with red and gray mottles

MAP UNIT COMPOSITION

Similar Soils

The well drained Noboco. Included similar soils make up less than 10 percent of the map unit.

Dissimilar Soils

These are the well drained Wagram and Bonneau soils

Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 4.0 to 6 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Erosion is the major management concern.

OCILLA SERIES

TAXONOMIC CLASSIFICATION:

Loamy, siliceous, thermic Aquic Arenic Paleudults.

REPRESENTATIVE PEDON

Ap -- 0 to 12 inches; brown (10YR 5/3) sand; weak fine granular structure; very friable; few fine roots; moderately acid; clear smooth boundary.

E1 -- 12 to 18 inches; light yellowish brown (10YR 6/4) loamy sand; weak fine subangular blocky structure; very friable; few fine roots; few fine and medium pores; moderately acid; clear wavy boundary.

E2 -- 18 to 25 inches; very pale brown (10YR 7/3) loamy sand; weak fine subangular blocky structure; very friable; few fine roots; few fine and medium pores; very strongly acid; clear wavy boundary.

Bt1 -- 25 to 48 inches; light yellowish brown (10YR 6/4) sandy clay loam; weak medium subangular blocky structure; friable; few fine roots; common fine and medium pores; few distinct clay films on faces of peds; few clean sand grains; common medium distinct strong brown (7.5YR 5/6) and common medium prominent yellowish red (5YR 5/6) masses of iron accumulation; common medium distinct gray (lOYR 6/1) iron depletions with clear boundaries in the matrix; very strongly acid; gradual wavy boundary.

Btg -- 40 to 60 inches; gray (lOYR 6/1) sandy clay loam; moderate medium subangular blocky structure; friable; common fine and medium pores; few distinct clay films on faces of peds; yellowish brown (lOYR 5/8) and strong brown (7.5YR 5/6) masses of iron accumulation; very strongly acid .

RANGE IN CHARACTERISTICS

Solum thickness is greater than 60 inches. Reaction is very strongly acid or strongly acid throughout . Iron depletions are **within** 12 to 30 inches of the surface.

The A or Ap horizon has hue of lOYR, value of 3 to 5, and chrow.a of 1 or 2. It is sand.

The E horizon has hue of lOYR, value of 4 to 8 and chroma of 1 to 4. It is sand.

The Bt horizon has hue of lOYR, value of 5 to 7 and chroma of 1 to 8. Masses of iron accumulation **in** shades of red, yellow or brown range from few to common. It is sandy loam or sandy clay loam.

The Btg horizon has hue of lOYR, value of 5 to 7 and chroma of 1 or 2. Masses of iron accumulation in shades of red, yellow, or brown range from few to common. It is sandy loam or sandy clay loam.

GEOGRAPHICALLY ASSOCIATED SOILS

These are the Blanton, Bonneau, Coxville, Noboco, Norfolk, Rains and Uchee soils. Blanton soils are grossarenic. Coxville and Rains soils have gray horizons immediately below surface layer. Bonneau soils do not have mottles of chroma 2 or less within 30 inches of the surface. Norfolk and Noboco soils do not have arenic epipedons.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Ocilla sand, 0 to 3 percent slopes, is located 0.5 mile north on paved road from the entrance of the research center, 0.5 mile north on paved road; 0.3 mile west on unpaved road; 0.1 mile west on unpaved road; 100 feet east.

MAP UNIT DESCRIPTION

OcB -- Ocilla loamy sand, 0 to *3 percent slopes*

SETTING

This map unit consists of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level to gently sloping low ridges, side slopes and around the inner rim of oval shaped depressions on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 12 inches brownish loamy sand

SUBSURFACE LAYER:

12 to 25 inches -- brownish loamy sand

SUBSOIL:

25 to 48 inches -- brownish sandy clay loam with red and gray mottles

48 to 60 inches -- mottled brownish, reddish and grayish sandy clay loam

MAP UNIT COMPOSITION

Similar Soils

These are the moderately well drained Goldsboro and Lynchburg and Rains soils. Included similar soils make up about 10 percent of the map unit.

Dissimilar Soils

These are the well drained Bonneau and Noboco soils. Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 1.0 to 2.5 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: moderate

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Wetness, low nutrient holding capacity, and soil blowing are the major management concerns.

RAINS SERIES

TAXONOMIC CLASSIFICATION:

Fine-loamy, siliceous, thermic Typic Paleaquults

Ap -- 0 to 8 inches; very dark gray (lOYR 3/1) sandy loam; weak fine granular structure; friable; common very fine and fine roots; few fine pores; few clean sand grains; slightly acid; abrupt wavy boundary.

Btg -- 8 to 14 inches; light brownish gray (lOYR 6/2) sandy clay loam; weak medium subangular blocky structure; friable; common very fine and fine roots; common distinct clay films on faces of peds; few clean sand grains; few fine distinct brownish yellow (lOYR 6/6) masses of iron accumulation; slightly acid; gradual wavy boundary.

Btg2 -- 14 to 28 inches; gray (lOYR 6/1) sandy clay loam; weak medium subangular blocky structure; friable; few fine roots; common fine and few coarse pores; common distinct clay films on faces of peds; common clean sand grains; common medium distinct reddish yellow (7.SYR 6/6) and few fine prominent red (2.SYR 5/8) masses of iron accumulation; strongly acid; gradual wavy boundary.

Btg3 -- 28 to 40 inches; gray (lOYR 6/1) sandy clay loam; weak medium subangular blocky structure; friable; common fine and medium pores; common distinct clay films on faces of peds; many coarse clean sand grains; common medium distinct yellowish brown (lOYR 6/6) and common medium prominent red (2.5YR 5/8) masses of iron accumulation; very strongly acid; abrupt wavy boundary.

Btg4 -- 40 to 51 inches; gray (lOYR 5/1) sandy clay; moderate medium subangular blocky structure; firm; many distinct clay films on faces of peds; common clean sand grains; common fine distinct light red (2.5YR 6/8) and olive yellow (2.5Y 6/6) masses of iron accumulation; very strongly acid; gradual wavy boundary.

Btg5 -- 51 to 61 inches; gray (lOYR 6/1) sandy clay; weak medium subangular blocky structure; firm; few medium distinct olive yellow (2.5Y 6/6) masses of iron accumulation; few medium faint gray (N/6) clay depletions with gradual boundaries in the matrix; very strongly acid; gradual wavy boundary .

Btg6 -- 61 to 65 inches; gray (lOYR 5/1) sandy clay and sandy clay loam; moderate medium subangular blocky structure parting to weak medium prismatic; firm; few fine distinct olive yellow (2.5Y 6/6) and few fine distinct red (2.5YR 4/6) masses of iron accumulation with clear boundaries in the matrix; very strongly acid; gradual wavy boundary.

BCg -- 65 to 70 inches; gray (lOYR 6/1) sandy clay loam and sandy clay; massive; firm; few medium distinct yellowish red (5YR 5/8) masses of iron accumulation; very strongly acid.

RANGE IN CHARACTERISTIC

Solum thickness is greater than 80 inches. Reaction is extremely acid to strongly acid throughout .

The Ap horizon has hue of lOYR, value of 2 to 5, and chroma of 0 to 2. It is sandy loam.

The Btg horizon has hue of lOYR, value of 4 to 7, and chroma of 0 to 2 . Masses of iron accumulation in shades of red, yellow or brown range from few to common. It is sandy loam, sandy clay loam, or sandy clay.

GEOGRAPHICALLY ASSOCIATED SOILS

These are the Bonneau, Coxville, Goldsboro, Lynchburg, Noboco and Ocilla soils. Bonneau, Goldsboro, Lynchburg, Noboco and Ocilla. In addition, Bonneau and Ocilla soils have A and E horizons more than 20 inches thick. Coxville soils have a clayey particle-size control section.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Rains sandy loam is located 0.4 mile northeast from the entrance of the research center; 0.4 mile north; .05 north on farm road; .05 mile east .

MAP UNIT DESCRIPTION

Ra -- Rains sandy loam.

SETTING

The Rains soils consists of very deep, poorly drained, moderately permeable soils that formed in loamy marine sediments. They are on nearly level flats or depression on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 8 inches grayish sandy loam.

SUBSOIL:

8 to 14 inches -- grayish sandy clay loam with yellowish mottles

14 to 40 inches -- grayish sandy clay loam with yellowish and reddish mottles

40 to 65 inches - grayish sandy clay with yellowish and reddish mottles

65 to 70 inches - grayish sandy clay loam and sandy clay with reddish mottles

MAP UNIT COMPOSITION

Similar Soils

These are the Coxville, Ocilla, and Lynchburg soils. Included similar soils make up less than 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: 0 to 1.0 feet

AVAILABLE WATER CAPACITY: moderate

EROSION HAZARD: slight

SURFACE RUNOFF: slow

ORGANIC MATTER CONTENT: moderate

AGRONOMIC INFORMATION:

Wetness is the major management concern.

UCHEE SERIES

TAXONOMIC CLASSIFICATION:

loamy, siliceous, thermic, Arenic Hapludults

REPRESENTATIVE PEDON

Ap -- 0 to 5 inch; grayish brown (lOYR 4/2) sand; weak fine granular structure; very friable; many medium, fine and very fine roots; common clean sand grains; slightly acid; clear wavy boundary .

E2 -- 5 to 25 inches; brownish yellow (lOYR 6/6) sand; weak fine subangular blocky structure; very friable; many fine roots; strongly acid; gradual wavy boundary .

Btl -- 25 to 34 inches; reddish yellow (7.5YR 6/8) sandy clay loam; weak medium subangular blocky structure; friable; few distinct clay films on faces of peds and along old root channels; few clean sand grains; common medium faint brownish yellow (7.5YR 6/6) masses of iron accumulation; very strongly acid; gradual wavy boundary .

Bt2 -- 34 to 41 inches; strong brown (7.5YR 5/6) sandy clay; common medium distinct brownish yellow (lOYR 6/8) and common medium prominent red (2.5YR 4/8) mottles;

moderate medium subangular blocky structure; firm in place; dense; few very fine roots; few distinct clay films on faces of peds; common clean sand grains; very strongly acid; gradual wavy boundary.

BC -- 41 to 46 inches; variegated color pattern consisting of 60 percent red (2.SYR 4/8) 40 percent brownish yellow (lOYR 6/8) sandy loam and sandy clay loam; massive; dense; very strongly acid; clear wavy boundary .

C -- 46 to 60 inches; variegated color pattern consisting of 34 percent (2.SYR 4/8), 33 percent white (lOYR 8/1), and 33 percent brownish yellow (lOYR 6/8) sandy loam and loamy sand; massive; common small masses of white (lOYR 8/1) balls of kaolin; very friable; very strongly acid.

RANGE IN CHARACTERISTICS

Solum thickness range from 40 to more than 60 inches . Reaction is very strongly acid or strongly acid throughout .

The A horizon has hue of lOYR, value of 4 to 5, and chroma of 1 to 4 . It is sand .

The E horizon has hue of lOYR, value of 5 to 6, and chroma of 4 to 6. It is sand.

The upper Bt horizon has hue of 2.SYR to lOYR, value of 5 to 7, and chroma of 4 to 8. Masses of iron accumulation in shades of red, yellow, or brown range from few to common. It is sandy loam or sandy clay loam.

The lower Bt horizon has hue of 7.SYR or lOYR, value of 5 to 7 and chroma of 4 to 8. Masses of iron accumulation in shades of red, yellow, or brown range from few to many. Iron depletions are within 40 to 60 inches of the surface. It is sandy clay loam, clay loam, sandy clay, or clay.

The BC horizon has hue of 7.SYR or lOYR, value of 5 to 7 and chroma of 4 to 8. Masses of iron accumulation in shades of red, yellow, or brown range from few to many. It is sandy loam or sandy clay loam.

The C horizon is variegated in shades of red, yellow, brown, gray, or white. It is sandy loam or sandy clay loam, or is stratified in coarser and finer textured material.

GEOGRAPHICALLY ASSOCIATED SOILS

These are the Bonneau, Blanton, Noboco and Wagram soils. Blanton soils are grossarenic. Coxville and Rains soils have gray horizons immediately below surface layer. Noboco soils do not have surface horizons more than 20 inches deep. Wagram and Bonneau soils are Paleudults.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Uchee sand, 0 to 3 percent slopes, is located 1.2 miles northeast from the entrance of the research center, 0.05 mile west on farm road; 250 southwest .

MAP UNIT DESCRIPTION

23B -- Uchee sand, 0 to *6 percent slopes.*

SETTING

This map unit consists of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridges and sloping side slopes near drainageways on the Coastal Plain.

REPRESENTATIVE PEDON

SURFACE LAYER:

0 to 5 inches grayish sand

SUBSURFACE LAYER:

5 to 25 inches brownish sand

SUBSOIL:

25 to 34 inches reddish sandy clay loam

34 to 41 inches brownish sandy clay with yellowish and reddish mottles

41 to 46 inches -- reddish and yellowish sandy clay loam and sandy loam

SUBSTRATUM:

46 to 60 inches -- mottled reddish, yellowish and white sandy loam and loamy sand

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Bonneau and Wagram soils, and the well drained to somewhat excessively drained Blanton soils. Included similar soils make up less than 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderately slow

SEASONAL HIGH WATER TABLE DEPTH: 3.5 to 5.0 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: slight

SURFACE RUNOFF: moderate

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION:

Droughtiness, low nutrient holding capacity, and soil blowing are the major management concerns.

WAGRAM SERIES

TAXONOMIC CLASSIFICATION:

Loamy, siliceous, thermic Typic Kandiudults

REPRESENTATIVE PEDON

Ap -- 0 to 8 inches; grayish brown (lOYR 5/2) sand; weak fine granular structure; very friable; few fine roots; slightly acid; clear wavy boundary.

E -- 8 to 25 inches; light yellowish brown (lOYR 6/4) sand; weak fine granular structure; very friable; few fine roots; slightly acid; clear wavy boundary.

Btl -- 25 to 40 inches; reddish yellow (7.5YR 6/8) sandy clay loam; weak fine subangular blocky structure; friable; few fine roots; common fine and medium pores; few distinct clay films on faces of peds; common medium faint brownish yellow (lOYR 6/6) masses of iron accumulation; strongly acid; gradual wavy boundary.

Bt2 -- 40 to 60 inches; reddish yellow (7.5YR 6/8) sandy clay loam; moderate medium subangular blocky structure; friable; few distinct clay films on faces of peds; common medium clean sand grains; few medium distinct yellowish red (5YR 5/6) masses of iron accumulation; strongly acid.

RANGE IN CHARACTERISTICS:

Solum thickness exceeds 60 inches. Reaction is strongly acid or very strongly acid throughout.

The A or Ap horizon has hue of lOYR, value of 5, and chroma of 2 to 4. Texture is sand.

The E horizon has hue of lOYR, value of 5 and chroma of 3 to 6. It is sand.

The Bt horizon has hue of 7.5YR or lOYR, value of 5 or 6, and chroma of 6 or 8. It is sandy loam or sandy clay loam. Masses of iron accumulation in shades of red, yellow or brown range from few to common. Iron depletions are 60 inches or more below the surface.

GEOGRAPHICALLY ASSOCIATED SOILS:

These are the Blanton, Bonneau, Noboco, Norfolk, and Uchee soils. Blanton soils are grossarenic. Bonneau soils have chroma 2 iron depletions in the matrix within 60 inches of the surface. Noboco and Norfolk soils do not have surface horizons more than 20 inches deep. Uchee soils have clay percentages that decrease by more than 20 percent of the maximum within 60 inches.

REPRESENTATIVE PEDON LOCATION:

A typical pedon of Wagram Sand, 0 to 6 percent slopes; is located 0.8 mile northeast of the research center entrance; 0.1 mile north on farm road; 0.1 mile east on farm road; 10 feet north.

MAP UNIT DESCRIPTION

JOB -- *Wagram loamy sand,* 0 to *4 percent slopes*

SETTING

The Wagram series consists of very deep, well drained, moderately permeable soils that formed in loamy marine sediments. These soils are on nearly level ridge tops and gently sloping areas on the Coastal Plain.

REPRESENTATIVE PROFILE:

SURFACE LAYER:

0 to 8 inches -- grayish brown sand

SUBSURFACE LAYER:

8 to 25 inches light yellowish brown sand

SUBSOIL:

25 to 40 inches reddish yellow sandy clay loam

40 to 60 inches reddish yellow sandy clay loam with yellowish red mottles

MAP UNIT COMPOSITION

Similar Soils

These are the well drained Bonneau and Uchee soils and the somewhat excessively drained Blanton soils. Included similar soils make 10 percent of the map unit.

Dissimilar Soils

These are the well drained Noboco and Norfolk soils.

Included dissimilar soils make up 0 to 10 percent of the map unit.

SOIL PROPERTIES:

PERMEABILITY: moderate

SEASONAL HIGH WATER TABLE DEPTH: greater than 6 feet

AVAILABLE WATER CAPACITY: low

EROSION HAZARD: slight *SURFACE RUNOFF:* slow

ORGANIC MATTER CONTENT: low

AGRONOMIC INFORMATION: Draughtiness, low nutrient holding capacity, and soil blowing are the major management concerns.